

Definition: Random Definitions

- $\text{pr}_i: X_1 \times \dots \times X_n \rightarrow X_i, (x_1, \dots, x_n) \mapsto x_i$
- The **power set** of X is $\mathcal{P}(X) = \{U: U \subseteq X\}$
- A subset of $\mathcal{P}$ is a **system of subsets** of X .
- $\emptyset$ is understood to be the linear combination of no vectors.
- A (an operator/mat/etc) is **nilpotent** w/ **degree** k if $A^k = 0, k \in \mathbb{N}$.
- For a set X and $f: X \rightarrow X, Y \subseteq X$ is **stable** under f if $f(Y) \subseteq Y$.
- $\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n, n \in \mathbb{N}$ is a **linear combination** of $\vec{v}_1, \dots, \vec{v}_n$.

Theorem: Random Theorems

- Group axioms for $(G, *)$: G closed under $*$; $*$ associative $[g*(h*k) = (g*h)*k]$, $\exists$ unique identity $\text{id} \in G$; $\exists$ unique inverse $g^{-1} \in G \forall g$.
- For G, H grps, $f: G \rightarrow H$ is a group hom. if $f(ab) = (f(a)f(b))$.
- For f a group hom. $\ker(f) = \{a \in G: f(a) = e\}$ is a normal subgroup of G .
- O-S: For G a group acting on a finite set $X, |\text{Orb}(x)| = |G|/|\text{Stab}(x)|$.
- Lagrange's Theorem: For G a finite group and $H \leq G$ a normal subgroup (invariant under conjugation, $ghg^{-1} \in H \forall g \in G, h \in H$); $|G| = |H| |G/H|$

Remark: Notation

- VS = Vector Space. VsubS = Vector subspace. $F-VS = F$ - Vector Space
- 'lin-ind' = 'linearly independent' and 'lin-dep' = 'linearly dependent'
- 'almost all' means 'all but finitely many'
- 'n-z' = 'non-zero'

Definition: Systems Of Linear Equations

Let F be a field. We seek to solve the system of n linear equations in m unknowns, described more easily by the extended coefficient matrix.

$$\begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \dots + a_{1m}x_m = b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2m}x_m = b_2 \\ \vdots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nm}x_m = b_n \end{pmatrix} \iff \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1m} & b_1 \\ a_{21} & a_{22} & \dots & a_{2m} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nm} & b_n \end{pmatrix}$$

- The set $L \subseteq F^m$ of m -tuples satisfying the equations is the **solution set**.
- For a linsys with soln set $\neq \emptyset$, all solns are obtained by adding an arbitrary soln of the homogenised sys to a fixed soln of the sys componentwise.

Major Definition: Rings and Fields

A **ring** is a set R equipped with operations $+$, $\cdot$ such that:

- $(R, +)$ is an abelian group, $\text{id} = 0_R$
- $(R, \cdot)$ is a **monoid** (has closure, associativity, identity $= 1_R$)
- Distributivity**: For all $a, b, c \in F$:
 - $a \cdot (b+c) = (a \cdot b) + (a \cdot c)$
 - $(a+b) \cdot c = (a \cdot c) + (b \cdot c)$

For both rings and fields:

- $(-a)b = -(ab) = a(-b)$
- $(-a)(-b) = ab$
- $0a = 0 = a0$
- $0, 1$ distinct (except in zero ring)

Ring Specific Lemmas and Definitions:

- The **null ring** or **zero ring** is defined as a ring where R is a single element - i.e. $\{0\}$ where $0+0=0$ and $0 \times 0 = 0$
- A **commutative ring** is one where $a \cdot b = b \cdot a$ for all $a, b \in R$
- $(-a)(b) = -(ab) = a(-b)$
 - $(-a)(-b) = ab$
 - $m(a+b) = ma + mb$
 - $m(n+a) = mn + na$
- The integers mod m (or $\mathbb{Z}/m\mathbb{Z}$) is a ring made up of congruence classes.
- A field is a n -z comring less $\{0\}$, with multiplicative inverses
- For $m \in \mathbb{N}, \mathbb{Z}/m\mathbb{Z}$ is a field iff m is prime

Major Definition: Modules and Vector Spaces

A **left module** M over a **ring** R (or an **R -module**) is a pair: an abelian group $M = (M, +)$, and a mapping $R \times M \rightarrow M: (r, a) \mapsto ra$ s.t. $\forall r, s \in R$ and $a, b \in M$:

$$\begin{aligned} r(a+b) &= (ra) + (rb) \\ (r+s)a &= (ra) + (sa) \\ r(sa) &= (rs)a \\ 1_R a &= a \end{aligned}$$

A **vector space** V over a field F is an extension of a module but over a field instead, and using vectors: $V = (V, +)$. $F \times V \rightarrow V: (\lambda, \vec{v}) \mapsto \lambda \vec{v}$ s.t. $\forall \lambda, \mu \in F$ and $\vec{v}, \vec{w} \in V$:

$$\begin{aligned} \lambda(\vec{v} + \vec{w}) &= \lambda \vec{v} + \lambda \vec{w} \\ (\lambda + \mu) \vec{v} &= \lambda \vec{v} + \mu \vec{v} \\ \lambda(\mu \vec{v}) &= (\lambda \mu) \vec{v} \\ 1 \vec{v} &= \vec{v} \end{aligned}$$

and they satisfy the following lemmas (for both):

- $0_R a = 0_M$ for all $a \in M$ or $\vec{0} = \vec{0}$ for all $\vec{v} \in V$
- $r \cdot 0_M = 0_M$ for all $r \in R$ or $\lambda \vec{0} = \vec{0}$ for all $\lambda \in F$
- $(-r)a = r(-a) = -(ra) \forall r \in R, a \in M$ or $(-1)\vec{v} = -\vec{v} \forall \vec{v} \in V$

Vector Space Lemmas and Definitions

- For X a set and V an F -VS, the set $\text{Maps}(X, V)$ of mappings $X \rightarrow V$ is an F -VS (define $(f+g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda(f(x))$).
- Given F -VSs $V_1, \dots, V_n, V_1 \times \dots \times V_n$ is an F -VS if we define addition/mult by scalars componentwise. We denote this VS by $V_1 \oplus \dots \oplus V_n$ and name it the **external direct sum**. $F^n = F \oplus \dots \oplus F$ n times.
- Two VsubSs V_1, V_2 of a VS V are **complementary** if addition defines a bijection: $V_1 \times V_2 \rightarrow V$. An example of an ∞ -dim VS is the space of all infinite sequences $a_n \in F$. A nice endo. with this VS is 'shifting' all the indices of the seq.
- Other ex.s are the spaces of all cont./cont. on an interval funcns/polynomials
- For a set X , the set $\text{Maps}(X, F)$ equipped with pointwise addition and scalar multiplication is an F -VS.

Major Definition: Subspaces

A sub-thing is a smaller but self-contained version of a thing

Vector Subspace: A subset U of a VS V . Needs $\forall \vec{u}, \vec{v} \in U, \lambda \in F$:

- $0 \in U$
- $\vec{u} + \vec{v} \in U$
- $\lambda \vec{u} \in U$

Subring A subset R' of a ring R under the same operations as R . Needs $\forall a, b \in R'$:

- R' has a multiplicative identity
- $a - b \in R'$
- $a \cdot b \in R'$

Submodule: A subset M' of an R -mod M under the same operations as M . Needs $\forall a, b \in M', r \in R$:

- $0_M \in M'$
- $a - b \in M'$
- $ra \in M'$

The intersection of subspaces/rings/modules is a subspace/ring/module

Subspace Lemmas and Definitions

- The solution set of a linsys of eqns seen earlier is a vector space
- A subset is a **hyperplane** of a VS if it is a proper VsubS ($V_{\text{subS}} \neq V$) and together with one other $\vec{v}$ generates the whole VS.
- The image of a VsubS under a linear mapping is again a VsubS, and the preimage of a VsubS under a linear mapping is again a VsubS.

Subring Lemmas and Definitions

- If R' a subring of R ; $f(R')$ is a subring of S .
- $\text{im}(f)$ is a subring of S .
- Let $f(1_R) = 1_S$.

Then if x is a unit in R , $f(x)$ is a unit in S and $(f(x))^{-1} = f(x^{-1})$.

In this case f restricts to a group homomorphism $f|_{R^\times}: R^\times \rightarrow S^\times$.

Submodule Lemmas and Definitions

- $\ker f$ is a submodule of M and $\text{im} f$ is a submodule of N .
- Let R be a ring, M an R -module, and $T \subseteq M$. Then the **submodule of M generated by T** is the set $\langle T \rangle = \{r_1 t_1 + \dots + r_m t_m: t_1, \dots, t_m \in T, r_1, \dots, r_m \in R\}$ together with the zero element if $T = \emptyset$.
- M is **finitely generated** if it is generated by a finite set
- M is called **cyclic** if it is generated by a singleton
- $\langle T \rangle$ is the smallest submodule of M containing T , for $T \subseteq M$.
- If M_1, M_2 are submods of M , $M_1 + M_2 := \{a+b: a \in M_1, b \in M_2\}$ is too.

Major Definition: Homo no homo

Everything has its own homomorphism and they are all the same thing

Linear Mapping: Hom. on a VS. Needs $\forall \vec{u}, \vec{v} \in U, \lambda \in F$:

- $f(\vec{u} + \vec{v}) = f(\vec{u}) + f(\vec{v})$
- $f(\lambda \vec{u}) = \lambda f(\vec{u})$

Ring Hom.: Hom. on R -Hom: a ring. Needs $\forall x, y \in R$:

- $f(x+y) = f(x) + f(y)$
- $f(xy) = f(x)f(y)$

Hom. on an R -mod. Needs $\forall a, b \in M, r \in R$:

- $f(a+b) = f(a) + f(b)$
- $f(ra) = rf(a)$

- A bijective homomorphism is called an **isomorphism**
- Two objects with an iso. are called **isomorphic**, written $A \cong B$
- A homomorphism $V \rightarrow V$ is called an **endomorphism** of V
- An isomorphism $V \rightarrow V$ is called an **automorphism** of V
- The composition of two hom.s is a hom. in all cases
- The inverse of an iso. is an iso. in all cases

Image and Kernel

The image and kernel of a mapping $f: M \rightarrow N$ are:

- Image:** $\text{im} f = \{f(a): a \in M\} \subseteq N$
- Kernel:** $\ker f = \{a \in M: f(a) = 0_N\} \subseteq M$

Any hom. is injective if and only if it has kernel $0/\vec{0}$ /etc

Major Theorem: Properties of Linear Mappings

Let V, W be F -VSs and $B \subseteq V$ be a basis.

- The set of all hom.s
- $V \rightarrow W$ is $\text{Hom}_F(V, W) = \text{Hom}(V, W) \subseteq \text{Maps}(V, W)$, and is itself a VS
- The **dual space** of V is $V^* = \text{Hom}(V, F)$
- The auto.s of a V form a subgroup of its permutation group, the **general linear** or **automorphism** group $GL(V) = \text{Aut}(V) \subseteq \text{Maps}^\times(V, V)$.
- The image and preimage of a VS under a linear mapping are VSs.
- A hom. can be restricted to a bijec $\text{Hom}_F(V, W) \rightarrow \text{Maps}(B, W), f \mapsto f|_B$, ie a hom. determines/is determined by the values it takes on a basis.
- Every injective hom. $f: V \rightarrow W$ has a **left inverse** g s.t. $g \circ f = \text{id}_V$
- Every surjective hom. $f: V \rightarrow W$ has a **right inverse** g s.t. $f \circ g = \text{id}_W$
- For $n \geq 0 \in \mathbb{Z}$, a VS V over F is isomorphic to F^n iff $\dim V = n$.
- A point sent to itself by a hom. is a **fixed point** of the mapping. Given a mapping $f: X \rightarrow X$, the set of fixed points is $X^f = \{x \in X: f(x) = x\}$.
- Given VS V , VsubSs $V_1, \dots, V_n$, and linmaps $f_i: V_i \rightarrow V: f: V_1 \oplus \dots \oplus V_n \rightarrow V, f(v_1, \dots, v_n) = f_1(v_1) + \dots + f_n(v_n)$ is an isomorphism. We then write $V = V_1 \oplus V_2$ and say V is the **internal direct sum** of V_1, V_2 .

Major Theorem: Properties of Ring Hom.s

Let R and S be rings and $f: R \rightarrow S$ a ring hom. Then $\forall x, y \in R$ and $m \in \mathbb{Z}$ ($0_R, 0_S$ are the zeros of R, S):

- $f(0_R) = 0_S$
- $f(-x) = -f(x)$
- $f(x-y) = f(x) - f(y)$
- $f(mx) = mf(x)$
- $f(x^n) = (f(x))^n$ for all $x \in R$ and $n \in \mathbb{N}$
- For $x \in R, n \in \mathbb{N}, f(x^n) = (f(x))^n$

The **zero ring homomorphism** is $f: R \rightarrow S; r \mapsto 0$.

- $f(1_R)$ is not necessarily equal to 1_S .
- If S has no zero-divs then either f is the zero hom., or $f(1_R) = 1_S$.

Major Theorem: Rank-Nullity Theorem

Let V, W be VSs and $f: V \rightarrow W$ a linear mapping. Then $\dim W = \dim(\ker(f)) + \dim(\text{im}(f))$. $\dim(\text{im}(f))/\dim(\ker(f))$ are the **rank/nullity** of f .

Definition: Matrices

$A: \{1, \dots, n\} \times \{1, \dots, m\} \rightarrow Z$ is an $n \times m$ -matrix with coefficients in Z .

- The set of all $n \times m$ -matrices with coefficients in Z is denoted $\text{Mat}(n \times m; Z)$; $= \text{Maps}(1, \dots, n \times 1, \dots, m, Z)$
- If $m = n$ we have a **square matrix** and write $\text{Mat}(n, Z)$
- For square matrix A , its **trace** $\text{tr}(A)$ is the sum of its diagonal entries
- $P \in \text{Mat}(n, \mathbb{R})$ is **orthogonal** if $P^T P = I_n$, ie $P^T = P^{-1}$.
- $P \in \text{Mat}(n, \mathbb{C})$ is **unitary** if $\overline{P}^T P = I_n$, ie $\overline{P}^T = P^{-1}$.
- $P \in \text{Mat}(n, \mathbb{C})$ is **hermitian** if $P = \overline{P}^T$, and the mapping on the standard $\mathbb{R}^n$ described by P is self-adjoint.
- For $A \in \text{Mat}(n, \mathbb{C})$, $\exp(A) := \sum_{k=0}^{\infty} \frac{1}{k!} A^k$. For commutative A, B , $\exp(A+B) = \exp(A)\exp(B)$

Theorem: Properties Of Matrices

- If $A, A' \in \text{Mat}(n \times m; F), B, B' \in \text{Mat}(m \times \ell; F), C \in \text{Mat}(\ell \times k; F)$, then:
 - $(A + A')B = AB + A'B$
 - $(A(B + B')) = AB + AB'$
 - $(AB)C = A(BC)$
- A square A is **invertible** if $\exists A^{-1}$ of same size s.t. $AA^{-1} = A^{-1}A = I$
- An **elementary** matrix is one which differs from I in at most one entry
- Every $M \in \text{Mat}(n, F)$ can be written as a product of elementary matrices
- A matrix whose only non-zero entries lie on the diagonal, with first ones along the diagonal then zeroes, is in **Smith Normal Form**.
- For each $A \in \text{Mat}(n \times m; F), \exists$ invertible P, Q s.t. PAQ is in SNF.
- The **rank** of a matrix A is the dimension of the subspace of F^n generated by the columns or rows of A (the dim of each are always equal).
- A is of **full rank** if the rank is equal to the number of rows or columns
- $\text{rk}(A+B) \leq \text{rk}A + \text{rk}B$, and $\text{rk}(AB) \leq \min(\text{rk}A, \text{rk}B)$.
- For R a comring, $A, B \in \text{Mat}(n; R)$:
 - $\det(AB) = \det(A)\det(B)$
 - $\det(A^T) = \det(A)$
 - A is invertible iff $\det(A)$ is a unit in R , ie $\det(A) \in R^\times$
- For F a field, $A, B \in \text{Mat}(n; F)$:
 - $\det(A) \neq 0$ iff A is invertible
 - If A is invertible then $\det(A^{-1}) = \det(A)^{-1}$
 - $\det(A^{-1}BA) = \det(B)$ (the det is independent of the choice of basis)
- When performing Gaussian Elimination, row addition does not change the determinant, and row swaps change the sign only

Vector Spaces

Definition: The Free Vector Space

The **free vector space** on $X, F(X) \subseteq \text{Maps}(X, F)$, is the VsubS of all mappings which send almost all $x \in X$ to zero.

Theorem: 2x2 Matrix Inverse Formula

For a 2x2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ the inverse $A^{-1} = \frac{1}{\det A} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

Use this at your own risk - no idea where it holds beyond $\mathbb{R}$:))

Definition: Span and Generating Sets

Let T be a subset of an F -VS $V, (T)$, the **vector subspace generated by T** or **span** of T is the set of all linear combinations of vectors in T .

- $\langle \emptyset \rangle = \vec{0}$
- If $v \in \langle T \rangle, \langle T \cup \{v\} \rangle = \langle T \rangle$.
- A subset of V is a **generating/spanning set** if its span is V .
- A VS with a finite generating set is **finitely generated**.
- Given $V_1, \dots, V_n$ VsubSs of V , we define the sum $V_1 + \dots + V_n$ to be the subspace $\langle V_1 \cup \dots \cup V_n \rangle$ of V .
- If the hom. given by addition: $V_1 \times \dots \times V_n$ is also an injection the sum is **direct** and can be written $V_1 \oplus \dots \oplus V_n$.

Definition: Linear Independence

A subset L of a VS V is **linearly independent** if for all $\alpha_1, \dots, \alpha_n \in F$ and distinct $\vec{v}_1, \dots, \vec{v}_n \in L, \alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n = 0$ only if all $\alpha_i = 0$. Otherwise it is **linearly dependent**.

- The FELA: For VS $V, L \subseteq V$ lin-ind, and $E \subseteq V$ generating, $|L| \leq |E|$.

Definition: Bases

Let V be an F -VS. A **basis** of V is a lin-ind generating set in V .

- The family $(\vec{v}_i)_{i \in I} \subseteq V$ is a basis of V iff the 'evaluation mapping' $\Phi: F^I \rightarrow V; (\alpha_1, \dots, \alpha_n) \mapsto (\alpha_1 \vec{v}_1 + \dots + \alpha_n \vec{v}_n)$ is a bijection.
- TFAE for a subset E of a VS V :
 - E is a basis
 - E is minimal among generating sets ($\forall \vec{v} \in E, E \setminus \vec{v}$ is not generating)
 - E is maximal among lin-ind subsets ($\forall \vec{v} \in E, E \cup \vec{v}$ is lin-dep)
- A family $(\vec{v}_i)_{i \in I}$ is a basis of V iff for all $\vec{v} \in V$ there exists exactly one family $(a_i)_{i \in I}$ of $a_i \in F$ s.t. almost all $a_i = 0$, and $\vec{v} = \sum_{i \in I} a_i \vec{v}_i$.

Major Theorem: Cardinality Of Bases

Let V be a finite VS. Then V has a finite basis, cannot have an infinite basis, and any two bases of V have the same cardinality

Lemma: The Exchange Lemma, Steinitz

Let V be a VS, $E \subseteq V$ generating, $L \subseteq V$ and $M \subseteq E$ lin-ind, L finite:

- The Exchange Lemma: If $\vec{w} \in V \setminus M$ is s.t. $M \cup \{\vec{w}\}$ is lin-ind, there exists $\vec{e} \in E \setminus M$ s.t. $(E \setminus \{\vec{e}\}) \cup \{\vec{w}\}$ is generating for V .

Steinitz Exchange Theorem: There exists an injection $\phi: L \hookrightarrow E$ s.t. $(E \setminus \phi(L)) \cup L$ is generating for V (apply Exchange Lemma repeatedly)

Definition: Dimensions Of Vector Spaces

Let V be a finitely-generated VS, $L \subseteq V$ lin-ind, $E \subseteq V$ generating. For a finitely-generated VS V , its **dimension** $\dim V$ is the cardinality of any basis of V . For dimension as an F -VS write $\dim_F V$. Also:

- $|L| \leq \dim V$, if $|L| = \dim V$ then L is a basis
- $|E| \geq \dim V$, if $|E| = \dim V$ then E is a basis
- A proper VsubS of V has a strictly smaller dimension than V
- For U, V Vsubspaces of V , $\dim(U \cap W) + \dim(U \cup W) = \dim(U) + \dim(W)$.
- Given $V_1, \dots, V_n$ F -VSs, $\dim(V_1 \oplus \dots \oplus V_n) = \dim(V_1) + \dots + \dim(V_n)$

Major Theorem: Rank-Nullity Theorem

Let V, W be VSs and $f: V \rightarrow W$ a linear mapping. Then $\dim W = \dim(\ker(f)) + \dim(\text{im}(f))$. $\dim(\text{im}(f))/\dim(\ker(f))$ are the **rank/nullity** of f .

Linear Mappings and Matrices

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Theorem: Properties Of Matrices

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- Every $M \in \text{Mat}(n, F)$ can be written as a product of elementary matrices
- A matrix whose only non-zero entries lie on the diagonal, with first ones along the diagonal then zeroes, is in **Smith Normal Form**.
- For each $A \in \text{Mat}(n \times m; F), \exists$ invertible P, Q s.t. PAQ is in SNF.
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 - $\det(A) \neq 0$ iff A is invertible
 - If A is invertible then $\det(A^{-1}) = \det(A)^{-1}$
 - $\det(A^{-1}BA) = \det(B)$ (the det is independent of the choice of basis)
- When performing Gaussian Elimination, row addition does not change the determinant, and row swaps change the sign only

Theorem: Linking Linear Mappings And Matrices

Let F be a field and $m, n \in \mathbb{N}$. There is a bijection M between the space of hom.s $F^m \rightarrow F^n$ and $\text{Mat}(n \times m; F)$

Theorem: Properties Of Ideals

- $\ker f$ is an ideal of R .
- The intersection of any collection of ideals is an ideal.
- For I, J ideals of R ; $I + J := \{a+b : a \in I, b \in J\}$ is an ideal.
- For R a comring and $T \subseteq R$, the ideal I of R generated by T is the set $R(T) = \{r_1 t_1 + \dots + r_m t_m : t_1, \dots, t_m \in T, r_1, \dots, r_m \in R\}$ together with 0_R if $T = \emptyset$.
- If T is finite can write $R(t_1, \dots, t_n)$.
- For $m \in \mathbb{Z}$, $\mathbb{Z}(m) = m\mathbb{Z}$.
- For R a comring and $T \subseteq R$, $R(T)$ is the smallest ideal of R containing T .
- For R a comring, an ideal I is a **principal ideal** if $I = (t)$ for some $t \in R$.
- In a comring, $\{0\} = (0)$ and $R = (1_R)$ are always principal ideals.

Definition: Equivalence and Equivalence Relations

A relation R on a set X is a subset $R \subseteq X \times X$.
Only in this context, $(x, y) \in R$ is written xRy .
 R is an **equivalence relation** on X when for all $x, y, z \in X$:
(i) Reflexivity: xRx
(ii) Symmetry: xRy if and only if yRx
(iii) Transitivity: $(xRy \text{ and } yRz) \implies xRz$
* For $\sim$ an equivalence relation on X and $x \in X$; the set $E(x) := \{z \in X : z \sim x\}$ is the **equivalence class** of x .
 $E \subseteq X$ is an **equivalence class** for $\sim$ if $\exists x \in X$ s.t. $E = E(x)$.
• An element of an equivalence class is a **representative** of the class.
 $Z \subseteq X$ is a **system of representatives** for $\sim$ if it contains precisely one element from each equivalence class of $\sim$.
• For $x, y \in X$ TFAE: $x \sim y$; $E(x) = E(y)$; $E(x) \cap E(y) \neq \emptyset$
• **The set of equiv classes of X is $(X/\sim) := \{E(x) : x \in X\} \subseteq \mathcal{P}(X)$.**
• There is a surjective canonical mapping $\text{can} : X \rightarrow (X/d), x \mapsto E(x)$.

Definition: Quotient Vector Space

For V an F -VS, $W \subseteq V$ a VsubS, $\sim$ on V s.t. $x \sim y$ if $x - y \in W$, so for $x \in V$, $E(x) = W + x \subseteq V$. The **quotient VS** of V by W , $V/\sim = V/W$ is an F -VS with $\lambda E(x) = E(\lambda x)$ and $\text{can} : V \rightarrow V/W$ is surj. with $\ker = W$. If V finite-dim, so are W and V/W , and $\dim(V/W) = \dim(V) - \dim(W)$.

Thm: Universal Prop. Of The Set Of Equivalence Classes

Let $\sim$ be an equivrel on X .
If $f : X \rightarrow Z$ is a mapping s.t. $x \sim y \implies f(x) = f(y)$; there is a unique mapping $\bar{f} : (X/\sim) \rightarrow Z$, $\bar{f}(E(x)) := f(x)$, and $f = \bar{f} \circ \text{can}$.
Often when defining a map $g : (X/\sim) \rightarrow Z$, we define $f : X \rightarrow Z$ and say g is **well-defined** to indicate that $x \sim y \implies f(x) = f(y)$ as this guarantees the existence of $g = \bar{f}$.

Definition: Cosets and Factor Rings

Let R be a ring, $I \trianglelefteq R$ an ideal of R , and $\sim$ defined by $x \sim y \iff x - y \in I$.
 $x + I := \{x + i : i \in I\} \subseteq R$
is a **coset** of I in R or the **coset** of x w.r.t. I in R
The set R/I , the **factor ring** of R by I or **quotient** of R by I , is the set $(R/\sim)$ of cosets of I in R .
• R/I is a ring, with addition defined by $(x+I) + (y+I) = (x+y)+I$ and multiplication by $(x+I)(y+I) = xy+I$ for all $x, y \in R$.
• If R is commutative so is R/I .
• R/I is a nonzero ring if and only if $I \neq R$.
• If I is proper, and $r \in R^\times$; $r+I \in (R/I)^\times$ with $(r+I)^{-1} = r^{-1}+I$.

Theorem: Universal Property Of Factor Rings

Let R, S be rings and $I \trianglelefteq R$. Then:
1. The mapping $\text{can} : R \rightarrow R/I, r \mapsto r+I$ is a surjective ring hom. with kernel I .
2. For $f : R \rightarrow S$ a ring hom. with $f(I) = \{0_S\}$, ie $I \subseteq \ker f$, there exists a unique ring hom. $\bar{f} : R/I \rightarrow S$ s.t. $f = \bar{f} \circ \text{can}$.

Major Theorem: The First Isomorphism Theorem For Rings

Let R, S be rings. Then every ring hom. $f : R \rightarrow S$ induces a ring iso.: $\bar{f} : R/\ker f \xrightarrow{\sim} \text{im } f$

Definition: Factor Modules

Let R be a ring, M an R -module, and N a submodule of M .
For $a \in M$, the **coset** of a w.r.t. N in M is : $a+N = \{a+b : b \in N\}$.
• It is a coset of N in the abelian group M , so is an equiv class for the equivrel $a \sim b \iff a - b \in N$.
 M/N , the **factor/quotient** of M by N is the set $(M/\sim)$ of all cosets of N in M .
• This becomes an R -module by defining: $(a+N) + (b+N) := (a+b)+N$ and $r(a+N) = ra+N$, for all $a, b \in M, r \in R$.
• The R -module M/N is the **factor module** of M by the submodule N .

Theorem: Universal Property Of Factor Modules

Let R be a ring, L, M be R -modules, and N a submodule of M :
1. The mapping $\text{can} : M \rightarrow M/N, a \mapsto a+N$ for all $a \in M$ is a surjective R -hom. with kernel N .
2. If $f : M \rightarrow L$ is an R -hom with $f(N) = \{0_L\}$ so that $N \subseteq \ker f$, there exists unique hom. $\bar{f} : M/N \rightarrow L$ such that $f = \bar{f} \circ \text{can}$.

Major Theorem: First Isomorphism Theorem For Modules

Let R be a ring and M, N be R -modules.
Then every R -hom. $f : M \rightarrow N$ induces an R -iso. $\bar{f} : M/\ker f \xrightarrow{\sim} \text{im } f$.
• **2nd** Theorem: For K, L sub-mods of M ; L is a submod of $K+L = \{b+c : b \in K, c \in L\}$ and $K \cap L$ is a submod of K . Also, $\frac{K+L}{K \cap L} \cong \frac{K}{K \cap L} \oplus \frac{L}{K \cap L}$.
• **3rd** Theorem: Let K, L be sub-mods of M , with $L \subseteq K$. Then K/L is a submod of M/L and $\frac{M/L}{K/L} \cong M/K$.

Determinants And Eigenvalues

Definition: Permutations And Their Signs

The group (under composition) of all permutations of the set $\{1, \dots, n\}$ to itself, denoted $\mathfrak{S}_n$, is the n -**th symmetric group**. It has order $|\mathfrak{S}_n| = n!$.
• A **transposition** is a perm that swaps two elements (no effect on others).
• An **inversion** of a $\sigma \in \mathfrak{S}_n$ is a pair (i, j) s.t. $1 \leq i < j \leq n$ and $\sigma(i) > \sigma(j)$.
• The number of inversions of a σ is the **length** of σ , $\ell(\sigma)$.
• The **sign** of σ is the parity of the length of σ : $\text{sgn}(\sigma) = (-1)^{\ell(\sigma)}$.
• If $\text{sgn}(\sigma) = 1/-1$, σ is an **even/odd permutation**.
• $\text{sgn}(\sigma\tau) = \text{sgn}(\sigma)\text{sgn}(\tau)$.
• The set of even perms in $\mathfrak{S}_n$ forms a subgroup (it is the kernel of $\text{sgn}(\sigma)$, a grp hom.) of $\mathfrak{S}_n$ called the **alternating group** and denoted A_n .

Definition: Cofactors And Adjugate Matrices

Let $A \in \text{Mat}(n; R)$ for a comring R , $n, i, j \in \mathbb{N}$ s.t. $1 \leq i, j \leq n$. Then:
• The (i, j) **cofactor** of A is $C_{ij} = (-1)^{i+j} \det(A(i, j))$ where $A(i, j)$ is the matrix obtained by deleting the i^{th} row and j^{th} column of A .
• The **adjugate matrix** $\text{adj}(A)$ is the $(n \times n)$ matrix whose entries are $\text{adj}(A)_{ij} = C_{ji}$ (note order swap of i and j !).

Major Definition: The Determinant

Let R be a comring and $n \in \mathbb{N}$. The **determinant** is a mapping $\det : \text{Mat}(n; R) \rightarrow R$ given by: $\det(A) = \sum_{\sigma \in \mathfrak{S}_n} \text{sgn}(\sigma) a_{1\sigma(1)} \dots a_{n\sigma(n)}$. This is called the **Leibniz formula**. For $n=0$ $\det(A) = 1$.
Alternative characterisations are:
• Let F be a field. The mapping $\det : \text{Mat}(n; F) \rightarrow F$ is the unique alternating multilinear form on n -tuples of column vectors with values in F that takes the value 1_F on the identity matrix.
• Laplace Expansions: Let R be a comring and $A = (a_{ij}) \in \text{Mat}(n; R)$. For fixed i, j the i^{th} row/ j^{th} column expansion of the determinant is $\det(A) = \sum_{j=1}^n a_{ij} C_{ij} / \sum_{i=1}^n a_{ij} C_{ij}$.

Definition: Bi/Multilinear Forms

Let U, V, W be F -VSs. A **bilinear form** on $U \times V$ with values in W is a mapping $H : U \times V \rightarrow W$ linear in both entries. Ie, $\forall u_1, u_2 \in U, v_1, v_2 \in V$:
• $H(\lambda u_1, v_1) = \lambda H(u_1, v_1)$ $H(u_1 + u_2, v_1) = H(u_1, v_1) + H(u_2, v_1)$
• $H(u_1, \lambda v_1) = \lambda H(u_1, v_1)$ $H(u_1, v_1 + v_2) = H(u_1, v_1) + H(u_1, v_2)$
For $U = V$, a bilinear form is **symmetric** if for all $u, v \in U$, $H(u, v) = H(v, u)$, and it is **alternating/antisymmetric** if for all $u \in U$, $H(u, u) = 0$.
• If H is antisymmetric, $H(u, v) = -H(v, u)$.
Multilinear forms are just an extension of bilinear forms to higher dimensions - H must be linear in each argument.
• A multilinear form is **alternating** if $\exists i \neq j$ with $v_i = v_j \implies H(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = 0$ and if so, has the property $H(v_1, \dots, v_i, \dots, v_j, \dots, v_n) = -H(v_1, \dots, v_j, \dots, v_i, \dots, v_n)$ for all $v_1, \dots, v_n \in V$.
• Thus for any $\sigma \in \mathfrak{S}_n$, $H(v_{\sigma(1)}, \dots, v_{\sigma(n)}) = \text{sgn}(\sigma) H(v_1, \dots, v_n)$.

Theorem: Cramer's Rule and Jacobi

Cramer: Let R be a comring, $A \in \text{Mat}(n; R)$. Then $A \cdot \text{adj } A = (\det(A)) I_n$
Jacobi: For $A = (a_{ij})$, $a_{ij} = a_{ij}(t)$ funcs of t ; $\frac{d}{dt} \det(A) = \text{tr } \text{adj } A \frac{dA}{dt}$

Major Definition: Eigenvalues and Eigenvectors

Let $f : V \rightarrow V$ be an endo. of an F -VS V .
 $\lambda \in F$ is an **eigenvalue** of f iff there exists non-z $\vec{v} \in V$ s.t. $f(\vec{v}) = \lambda \vec{v}$.
Each such $\vec{v} \in V$ is called an **eigenvector** of f with **eigenvalue** λ .
 $E(\lambda, f)$, the **eigspace** of f with **eigval** λ , is $\{\vec{v} \in V : f(\vec{v}) = \lambda \vec{v}\}$.
• Each endo. of a n-z finite-dim VS over an algly-cld field has an eigval.
• For f with distinct eigenvals $\lambda_1, \dots, \lambda_n$, the eigvecs $\vec{v}_1, \dots, \vec{v}_n$ are lin-ind.

Definition: The Characteristic Polynomial

Let R be a comring and $A \in \text{Mat}(n; R)$. The polynomial $\chi_A(x) := \det(xI_n - A) \in R[x]$ is called the **characteristic polynomial** of A .
• For F a field, $A \in \text{Mat}(n; F)$; the eigvals of A are the roots of χ_A .

Theorem: Cayley-Hamilton Theorem

Let R be a comring and $A \in \text{Mat}(n; R)$. Then $\chi_A(x)$ evaluated at the matrix A , $\chi_A(A)$, equals zero.

Definition: Conjugacy

For R a comring, $A, B \in \text{Mat}(n; R)$:
 A and B are **conjugate** if $B = P^{-1}AP$ for invertible $P \in \text{GL}(n; R)$.
• Conjugacy is an equivalence relation on $\text{Mat}(n; R)$.
• The characteristic polynoms of conjugate matrices are the same

Definition: Triangularisability

Let $f : V \rightarrow V$ be an endo. of a finite-dim F -VS V . The following two statements are equivalent, and if they hold, f is **triangularisable**:
(1) $\chi_f(x)$, decomposes into linear factors in $F[x]$.
(2) V has an ordered basis $\mathcal{B} = (\vec{v}_1, \dots, \vec{v}_n)$ s.t. $f(\vec{v}_1) = a_{11}\vec{v}_1, f(\vec{v}_2) = a_{12}\vec{v}_1 + a_{22}\vec{v}_2, \dots, f(\vec{v}_n) = a_{1n}\vec{v}_1 + \dots + a_{nn}\vec{v}_n \in V$.
Equivalently, the matrix $\mathcal{B}[f]_{\mathcal{B}}$ is upper triangular.
• An endo. $A : F^n \rightarrow F^n$ is triangable iff A is conj to an upper triang mat B . Ie, any endo. of a finite-dim VS over an algly-cld field is triangable.

Definition: Diagonalisability

• An endo. $f : V \rightarrow V$ of an F -VS is **diagonalisable** iff there exists a basis of V made up of eigvecs of f .
• If V is finite-dim, this is equivalent to: $\exists$ ordered basis $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_n\}$ such that $\mathcal{B}[f]_{\mathcal{B}} = \text{diag}(\lambda_1, \dots, \lambda_n)$. Here $f(\vec{v}_i) = \lambda_i \vec{v}_i$.
• A square matrix $A \in \text{Mat}(n; F)$ is **diagonalisable** iff the linear mapping $F^n \rightarrow F^n$ given by left multiplication by A is diagonalisable.
That is, A is conjugate to a diagonal matrix ($\exists P \in \text{GL}(n; F)$ s.t. $P^{-1}AP = \text{diag}(\lambda_1, \dots, \lambda_n)$). In this case the columns of P are the vecs of a basis of F^n consisting of eigenvecs of A with eigenvals $\lambda_1, \dots, \lambda_n$.

Inner Product Spaces

Major Definition: Inner Products And Their Spaces

Let V be an $\mathbb{R}$ -VS. An **inner product** on V is a mapping $(-, -) : V \times V \rightarrow \mathbb{R}$ that satisfies, for all $\vec{x}, \vec{y}, \vec{z} \in V$ and $\lambda, \mu \in \mathbb{R}$:
(1) $(\lambda \vec{x} + \mu \vec{y}, \vec{z}) = \lambda(\vec{x}, \vec{z}) + \mu(\vec{y}, \vec{z})$
(2) $(\vec{x}, \vec{y}) = (\vec{y}, \vec{x})$ (together with (1) implies symmetric bilinear form)
(3) $(\vec{x}, \vec{x}) \geq 0$, = 0 iff $\vec{x} = 0$ (positive definite)
• The **standard inner product** on $V = \mathbb{R}^n$ is the dot product
• A **real inner product space** is a real VS endowed with an inner product.
For V a VS over $\mathbb{C}$ the definition is similar, swap $\mathbb{R}$ for $\mathbb{C}$ everywhere; condition (2) reads $(\vec{x}, \vec{y}) = \overline{(\vec{y}, \vec{x})}$; and the standard inner product is $(\vec{v}, \vec{w}) = v_1 \bar{w}_1 + \dots + v_n \bar{w}_n$. Note the SIP is not $\mathbb{C}$ -linear in the second variable.
• In a $\mathbb{R}/\mathbb{C}$ -VS, the **length/inner product norm/norm** is $\|\vec{v}\| := \sqrt{(\vec{v}, \vec{v})}$.
• Vectors of length 1 are **units**.
• $\vec{v}, \vec{w}$ are **orthogonal**, $\vec{v} \perp \vec{w}$, iff $(\vec{v}, \vec{w}) = 0$.
• Sets S, T are orthogonal, $S \perp T$, if $\vec{v} \perp \vec{w} \forall \vec{v} \in S, \vec{w} \in T$.
• An **orthonormal family** of vecs in an IPS is a $(\vec{v}_i)_{i \in I}$ s.t. all $\vec{v}_i$ have $\|\vec{v}_i\| = 1$ and are pairwise orthogonal (by Kronecker delta, $(\vec{v}_i, \vec{v}_j) = \delta_{ij}$).
• An orthonormal family that is a basis is an **orthonormal basis**. For V an IPS, $(\vec{v}_i)_{i \in I}$ an orthonormal basis; $\forall \vec{w} \in V$, $\vec{w} = \sum_{i \in I} (\vec{w}, \vec{v}_i) \vec{v}_i$.
• Every finite-dim IPS has an orthonormal basis

Dfn: Orthogonal Complements/Projections

Let V be an IPS, $T \subseteq V$, and U be a finite-dim subspace of V .
 $T^\perp = \{\vec{v} \in V : \vec{v} \perp \vec{t} \forall \vec{t} \in T\}$ is the **orthogonal** to T .
 $T^\perp$ is always a subspace of V even if T is not, and $T^\perp = \langle T \rangle^\perp$.
The space $U^\perp$ is the **orthogonal complement** to U .
• Every $\vec{v} \in V$ can be written as $\vec{v} = \vec{p} + \vec{r}$, $\vec{p} \in U$ and $\vec{r} \in U^\perp$.
• U and $U^\perp$ are complementary, ie $V = U \oplus U^\perp$.
The **orthogonal projection** from V onto U is $\pi_U : V \rightarrow V, \vec{v} = \vec{p} + \vec{r} \mapsto \vec{p}$.
• π_U is a linmap with $\text{im}(\pi_U) = U$ and $\ker(\pi_U) = U^\perp$.
• $\pi_U^2 = \pi_U$, ie π_U is idempotent
• Let $\{\vec{v}_1, \dots, \vec{v}_n\}$ be an orthonormal basis of U .
Then for all $\vec{v} \in V$, $\pi_U(\vec{v}) = \sum_{i=1}^n (\vec{v}, \vec{v}_i) \vec{v}_i$.

Major Theorem: The Cauchy-Schwarz Inequality

Let $\vec{v}, \vec{w}$ be vectors in an inner product space.
Then $|(\vec{v}, \vec{w})| \leq \|\vec{v}\| \|\vec{w}\|$ with equality iff $\vec{v}, \vec{w}$ are linearly dependent.
For V an IPS, $\vec{v}, \vec{w} \in V$, and λ a scalar (ie in the underpinning field):
• $\|\vec{v}\| \geq 0$, = 0 iff $\vec{v} = \vec{0}$.
• $\|\lambda \vec{v}\| = |\lambda| \|\vec{v}\|$.
• $\vec{v} + \vec{w} \leq \|\vec{v}\| + \|\vec{w}\|$.
• In a real IPS, $\exists$ unique $0 \leq \theta \leq \pi$ s.t. $\cos(\theta) = \frac{(\vec{v}, \vec{w})}{\|\vec{v}\| \|\vec{w}\|}$.

Theorem: Gram-Schmidt Process

Let $\vec{v}_1, \dots, \vec{v}_k$ be linearly ind. vectors in an IPS V . Then we can construct an orthonormal family $\vec{w}_1, \dots, \vec{w}_k$ such that for all $1 \leq i \leq k$, $\langle \vec{v}_1, \dots, \vec{v}_i \rangle = \langle \vec{w}_1, \dots, \vec{w}_i \rangle$. If V is finite-dim, we can use the G-S process to transform any basis into an orthonormal basis.
• For $k=1$ take $\vec{w}_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|}$ (normalise the length)
• Then given $\vec{w}_1, \dots, \vec{w}_k$; $\vec{w}_{k+1} = \frac{\vec{v}_{k+1} - \sum_{i=1}^k (\vec{v}_{k+1}, \vec{w}_i) \vec{w}_i}{\|\vec{v}_{k+1} - \sum_{i=1}^k (\vec{v}_{k+1}, \vec{w}_i) \vec{w}_i\|}$

Definition: Adjoints and Self-Adjointness

Let V be an IPS. Two endo.s $T, S : V \rightarrow V$ are **adjoint** if for all $\vec{f}, \vec{g} \in V$, $(T\vec{v}, \vec{w}) = (\vec{v}, S\vec{w})$. In this case $S = T^*$ and S is the **adjoint** of T .
• If V a finite-dim IPS and $T : V \rightarrow V$ an endo; T^* exists and is unique.
• An endo. of an IPS $T : V \rightarrow V$ is **self-adjoint** if $T = T^*$.
• Let $T : V \rightarrow V$ be a self-adjoint linmap on an IPS V . Then:
(1) Every eigenvalue of T is real
(2) If λ, μ are distinct eigenvals of T with eigenvecs $\vec{v}, \vec{w}$, then $(\vec{v}, \vec{w}) = 0$
(3) T has an eigenvalue.

Major Theorem: The Spectral Theorem

Let V be a finite-dim IPS and $T : V \rightarrow V$ be a self-adjoint linmap. Then V has an orthonormal basis consisting of eigenvectors of T .
• Let $A \in \text{Mat}(n, \mathbb{R})$ be symmetric. Then $\exists$ orthogonal $P \in \text{Mat}(n, \mathbb{R})$ such that $P^T A P = P^{-1} A P = \text{diag}(\lambda_1, \dots, \lambda_n)$, where the λ_i are the (necessarily real) eigvals of A repeated according to multiplicity.
• Let $A \in \text{Mat}(n, \mathbb{C})$ be hermitian. Then $\exists$ unitary $P \in \text{Mat}(n, \mathbb{C})$ s.t. $\bar{P}^T A P = P^{-1} A P = \text{diag}(\lambda_1, \dots, \lambda_n)$, where the λ_i are the (necessarily real) eigvals of A repeated according to multiplicity.

Jordan Normal Form

Definition: Jordan Blocks

For $r \in \mathbb{N}$: The **nilpotent Jordan block** of size r is the $(r \times r)$ matrix $J(r)$ defined by the rule $J(r)_{ij} = 1$ for $j = i+1$ and = 0 otherwise. The **Jordan block** of size r and **eigval** λ , for $\lambda \in F$, is the $(r \times r)$ matrix $J(r, \lambda)$ defined by the rule $J(r, \lambda) = \lambda I_r + J(r)$ with $DN = ND$.

Major Theorem: Jordan Normal Form

Let F be an algebraically closed field, V be a finite-dim VS, and $\phi : V \rightarrow V$ an endo. with characteristic polynomial $\chi_\phi(x) = (x - \lambda_1)^{a_1} \dots (x - \lambda_2)^{a_2} \dots (x - \lambda_s)^{a_s} \in F[x]$, $a_i \geq 1, \sum_{i=1}^s a_i = n$ for distinct $\lambda_1, \dots, \lambda_s \in F$. Then $\exists$ an ordered basis $\mathcal{B}$ of V s.t. the matrix of ϕ w.r.t. $\mathcal{B}$ is **diag** block diag with Jordan blocks on the diag. That is, $\mathcal{B}[\phi]_{\mathcal{B}} = \text{diag}(J(r_{11}, \lambda_1), \dots, J(r_{1m_1}, \lambda_1), J(r_{21}, \lambda_2), \dots, J(r_{sm_s}, \lambda_s))$ with $r_{11}, \dots, r_{1m_1}, r_{21}, \dots, r_{sm_s} \geq 1$ s.t. $a_i = r_{i1} + r_{i2} + \dots + r_{im_i}$, $(1 \leq i \leq s)$.

Theorem: Bézout's Identity

If $P_j(x) = \prod_{i=1, i \neq j}^s (x - \lambda_i)^{a_i}$, $\exists Q_j \in F[x]$ s.t. $\sum_{j=1}^s P_j(x) Q_j(x) = 1$.

Definition: Generalised Eigenspace

The **generalised eigenspace** of ϕ with eigval λ_i is $E^{\text{gen}}(\lambda_i, \phi) = \{\vec{v} \in V : (\phi - \lambda_i \text{id}_V)^{a_i}(\vec{v}) = \vec{0}\} \subseteq V$. For the **eigenspace** $E(\lambda_i, \phi) \subseteq E^{\text{gen}}(\lambda_i, \phi)$, remove the a_i term. $\dim(E^{\text{gen}}(\lambda_i, \phi))$ and $\dim(E(\lambda_i, \phi))$ are the **algebraic** and <